

# DC and FT occupancy for ALERT

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# Informations

Use gemc 5.10, latest version of the detector geometry from [github.com/mpaolone/detectors](https://github.com/mpaolone/detectors) .

Generate 1014357 electron at 10.6 GeV per event = 650 nA current in a 248.5 ns time windows (from Viktoriya Sergeyeva study).

$$N_{electrons} = I * 6.24 \times 10^{18} \text{ (charges/s)} * T_{window} \text{ (s)} = 1014357.$$

Use the helium 4 target.

Use inbending configuration (negative magnetic shield)

Use the “official” software for occupancy develop by R. De Vita  
[github.com/raffaelladevita/clas12-background](https://github.com/raffaelladevita/clas12-background)

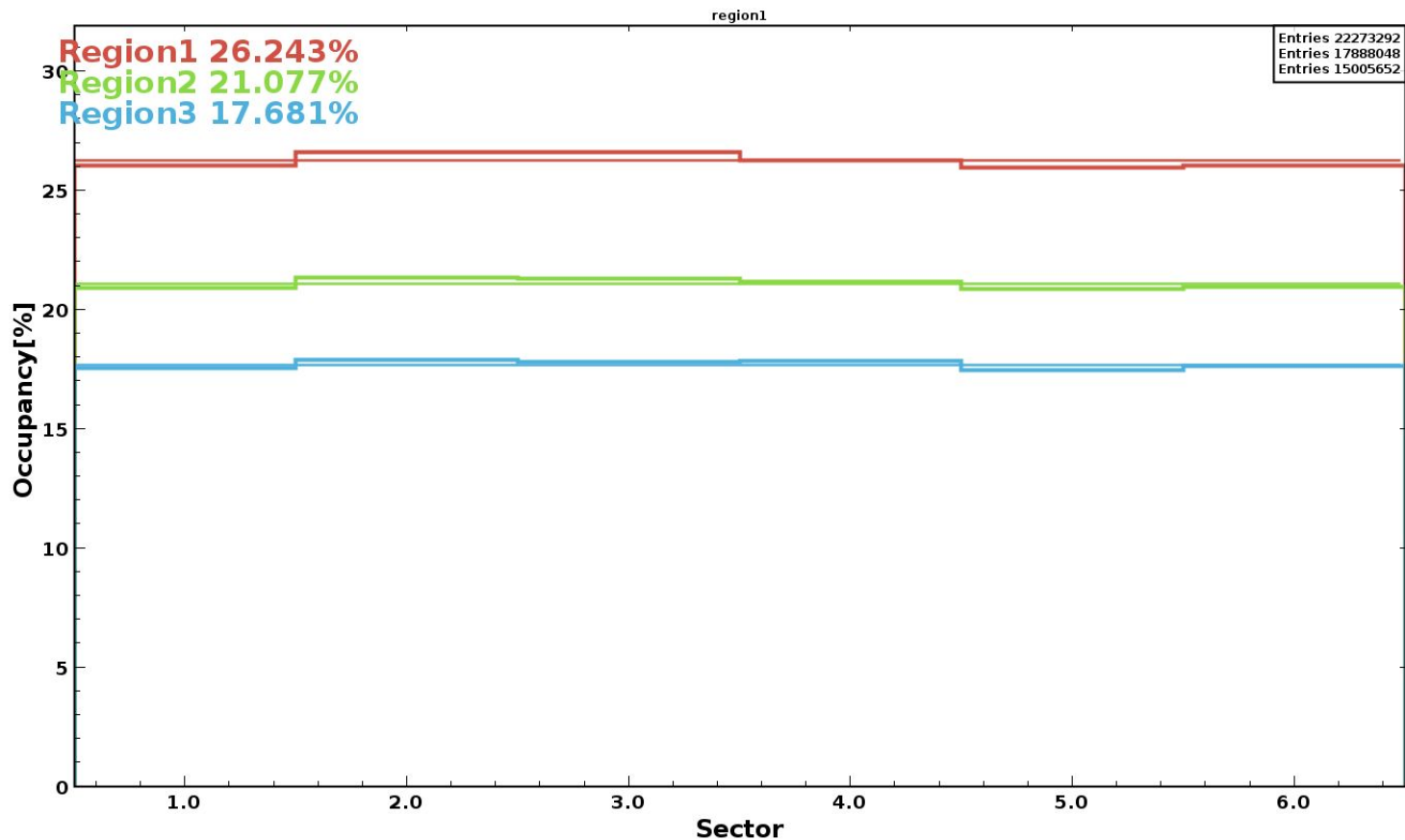
# Detectors

ALERT: atof, ahdc, external shell

CENTRAL : ctof (cad), cnd (-3cm shift)

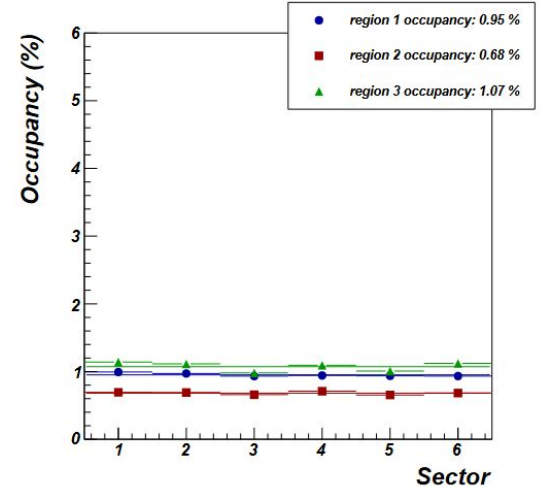
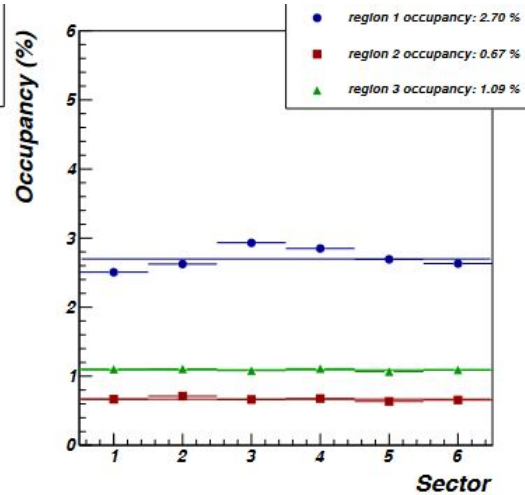
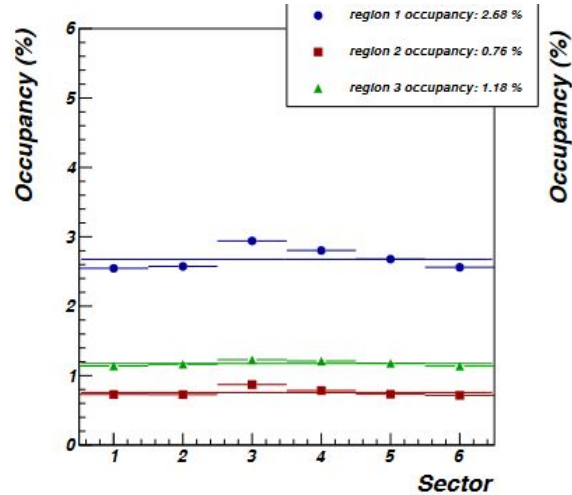
FORWARD : htcc (cad, -1.94cm shift), ft, beamline, dc, ftof, ec, pcal, ltcc, rich

# Results



30% occupancy.  
10 times the  
expected  
occupancy.  
Not sure why, still  
try to understand.

# Others results



Study from R. De Vita CLAS12 Note  
2017-016  
124,000 electrons going through the LH2  
target within a 250 ns

Study from Viktoriya Sergeyeva  
CLAS12 DC occupancy for  
ALERT luminosity and setup.  
Nelectrons = 1014357.

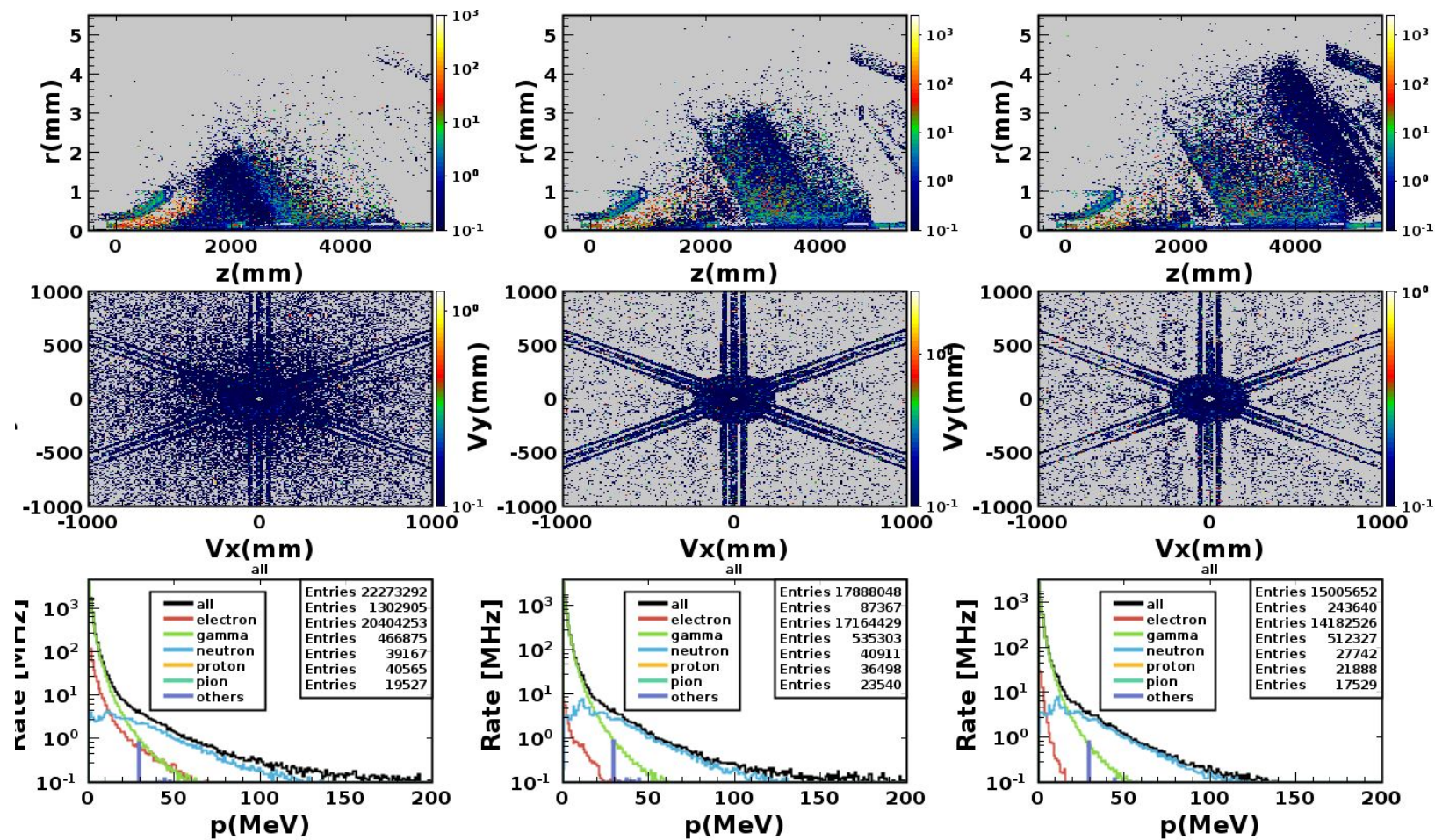
# Next steps

Understand the much bigger values from the DC occupancy.

FT occupancy : need to be implemented in the software. email from R. De Vita :  
“The next detector to be implemented will be FT.”

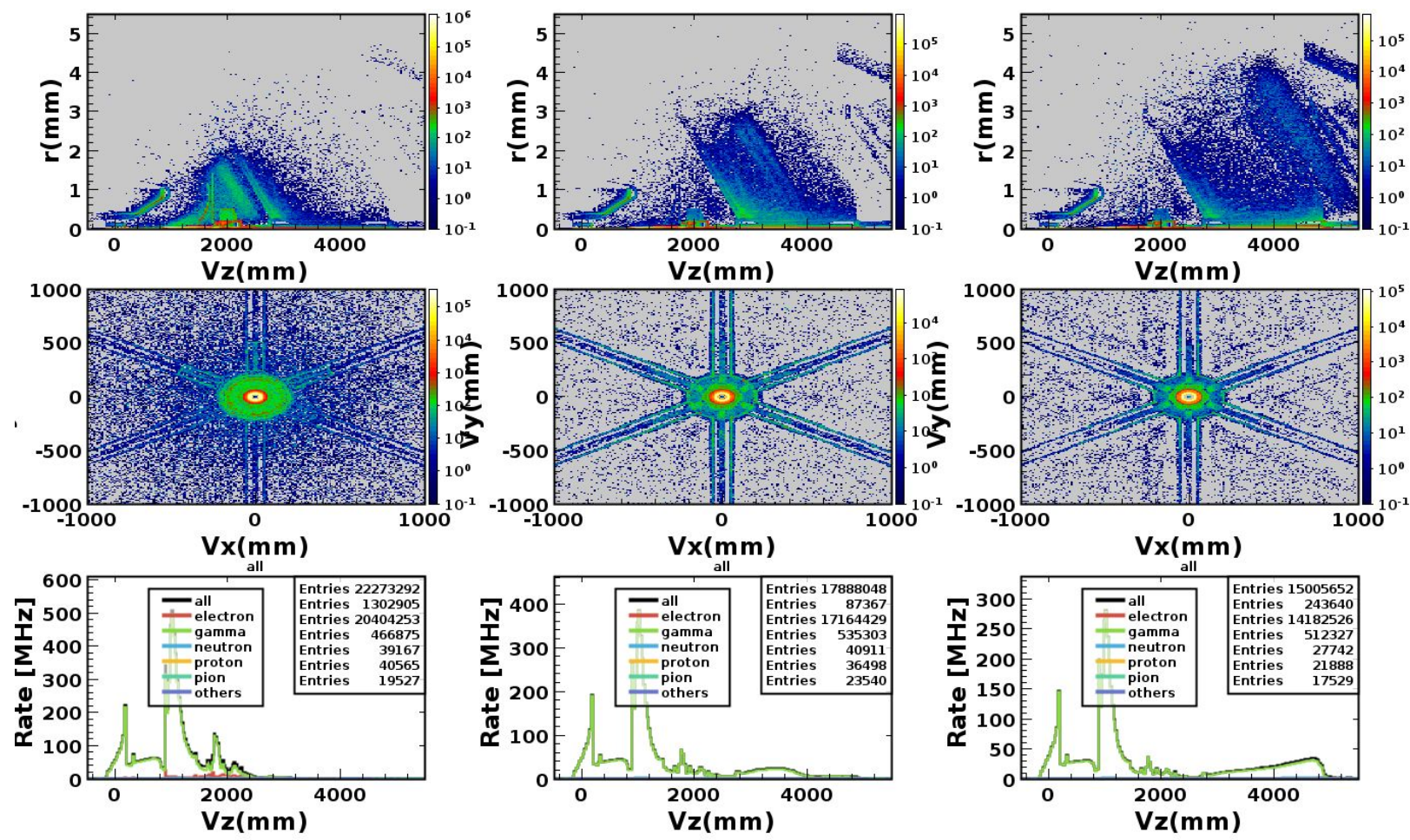
Test the occupancy software to see if the issues come from the software or gemc.

# Origin of Bg - Energy



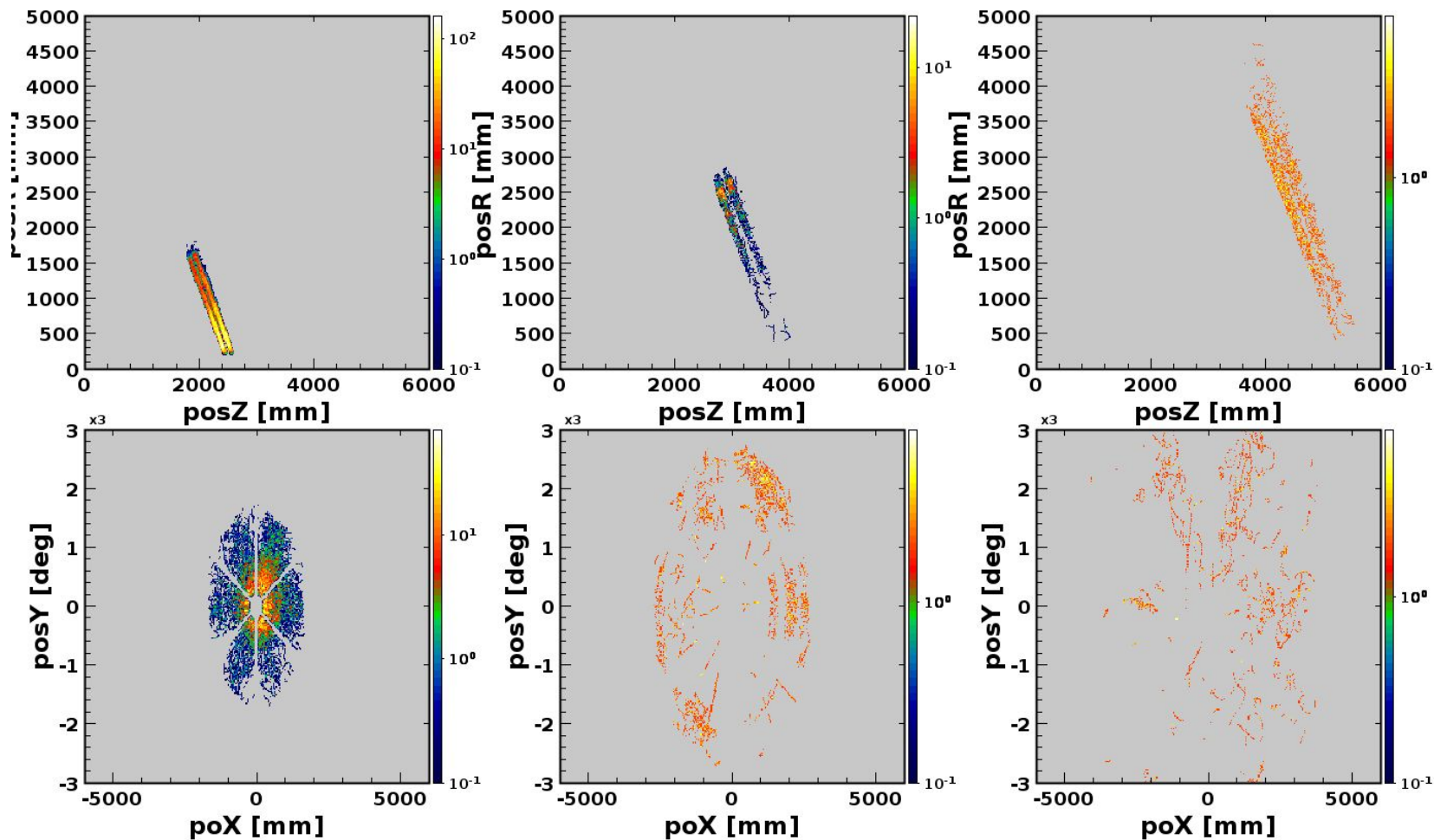


# Origin of Bg - Position

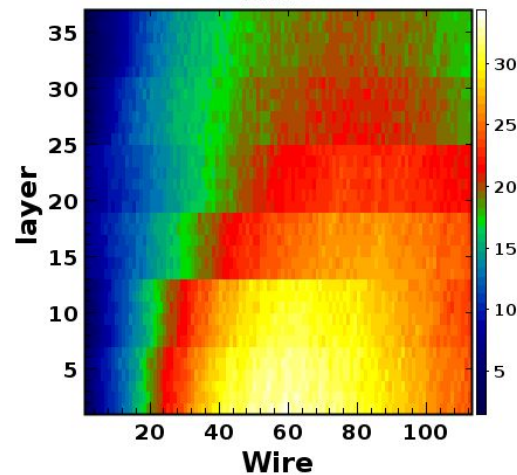
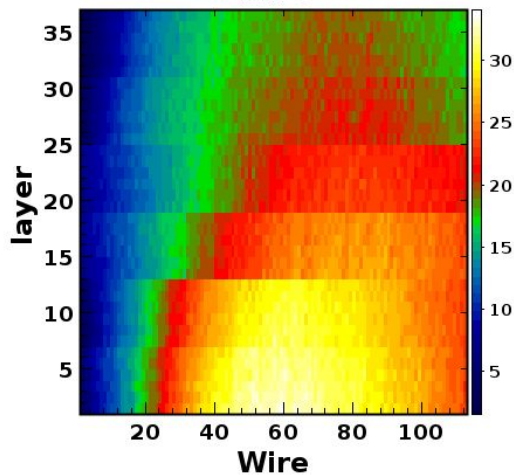
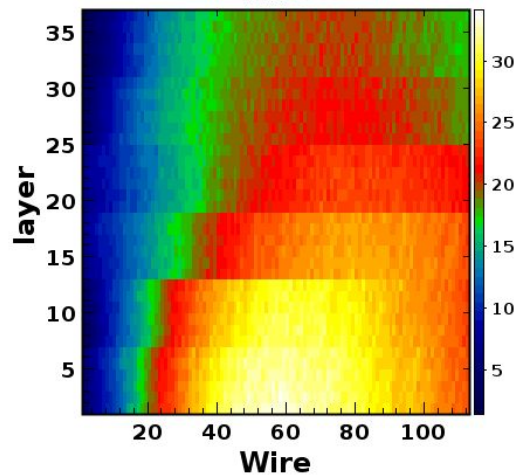
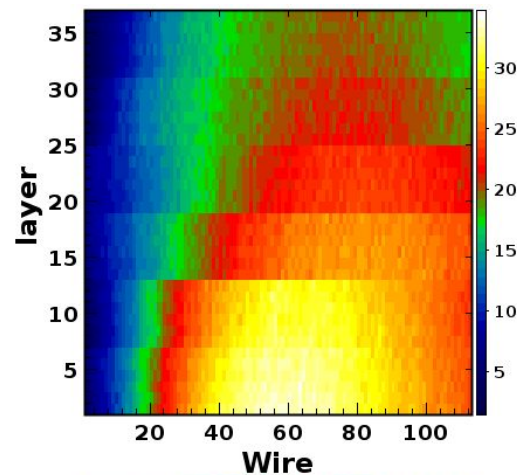
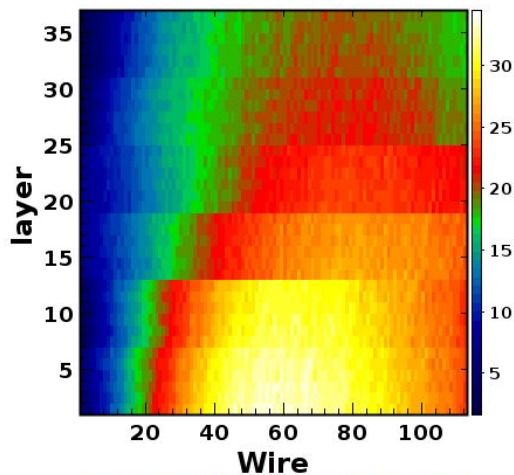
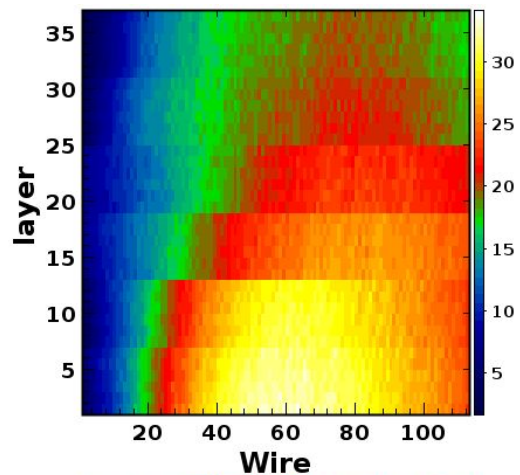




# Position of Bg



# DC sector occupancy



# Measurement uncertainty

the uncertainty in the estimated mean is equal to the uncertainty in each individual measurement divided by the square root of the number of times the measurement was repeated.

If you measure 5 times = divide the uncertainty by  $\sqrt{5} = 2.24$

If you measure 10 times = divide the uncertainty by  $\sqrt{10} = 3.16$

What will density be used for, to find out if this uncertainty is limiting us and how much we need to reduce it?